There is no known effective treatment for tardive dyskinesia, antiparkinson agents usually do not alleviate the symptoms of this syndrome. It is suggested that all antipsychotic agents be discontinued if these symptoms appear. Should it be necessary to reinstitute treatment, or increase the dosage of the agent or switch to a different antipsychotic agents, this syndrome may be masked. It has reported that fine vermicular movement of the tongue may be an early sign of the syndrome and if the medication is stopped at that time the syndrome may not develop.

Other CNS Effects- Insomnia, restlessness, anxiety, euphoria, agitation, drowsiness, depression, lethargy, headache, confusion, vertigo, grand mal seizures, exacerbation of psychotic symptoms including hallucinations and catatonic-like behavioral state which may be responsive to drug withdrawal and of treatment with anticholinergic drugs.

Body as a whole: as with other neuroleptic drugs, hyperpyrexia, that is rarely associated with rhabdomyolysis and acute renal failure, has occurred.

Cardiovascular Effects: Tachycardia and hypotension.

Hematologic Effects: Reports have appeared citing the occurrence of mild and usually transient leukpenia and leukocytosis, minimal decreases in red blood cell counts, anemia, or a tendency toward lymphomonocytosis. Agranulocytosis has rarely been reported to have occurred with the use of Haloperidol and then only in association with other medication.

Liver Effects: Impaired liver function and/or jaundice have been reported.

Dermatologic Reactions: Maculopapular and acneiform skin reactions and isolated cases of photosensitivity and loss of hair.

Endocrine Disorders: Lactation, breast engorgement, mastalgia, menstrual irregularities, gynecomastia, impotence, increased libido, hyperglycemia, hypoglycemia and hyponatremia.

Gastrointestinal Effects: Anorexia, Constipation, diarrhoea, hypersalivation, dyspepsia, nausea and vomiting.

Autonomic Reactions: Dry mouth, blurred vision, urinary retention and diaphoresis.

Respiratory Effects: Laryngospasm, bronchospasm and increased depth of respiration.

Other: Cases of sudden and unexpected death have been reported in association with the administration of Haloperidol.

The nature of the evidence makes it impossible to determine definitely what role, if and Haloperidol played in the outcome of the reported cases. The possibility that Haloperidol caused death cannot or course, be excluded, but it is to be kept in mind that sudden and unexpected death may occur in psychotic patients when they go untreated or when they are treated with other neuroleptic drugs.

Pregnancy and Lactation

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self- limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation.

Royce Haloperidol Tablet 1.5mg should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

The safety of haloperidol in pregnancy has not been established. There is some evidence of harmful effects in some, but not all animal studies. There have been a number of reports of birth defects following foetal exposure to haloperidol for which a causal role for haloperidol cannot be excluded. Reversible extrapyramidal symptoms have been observed in neonates exposed to haloperidol in utero during the last trimester of pregnancy. Haloperidol should be used during pregnancy only if the anticipated benefit outweighs the risk and the administered dose and duration of treatment should be as low and as short as possible.

Haloperidol is excreted in breast milk. There have been isolated cases of extrapyramidal symptoms in breast-fed children. If the use of Haloperidol is essential, the benefits of breast feeding should be balanced against its potential risks.

Symptoms and Treatment of Overdose

Manifestations: In general, the symptoms of overdosage would be an exaggeration of known

pharmacologic effects and adverse reactions, the most prominent of which would be:

i) Severe extrapyramidal reactions
ii) Hypotension
iii) Sedation

The patients would appear comatose with respiratory depression and hypotension which could be severe enough to produce a shock-like state. The extrapyramidal reaction would be manifest by muscular weakness or rigidity and a generalized or localized tremor as demonstrated by the akinetic or agitans types respectively. With accidental overdosage, hypotension rather than hypotension occurred in a two-year old child.

Treatment: Gastric lavage or induction of emesis should be carried immediately followed by administration of activated charcoal. Since there is no specific antidote, treatment is primarily supportive. A patent airway must be established by use of an oropharyngeal airway or endotracheal tube or, in prolonged cases of coma, by tracheotomy. Respiratory depression may be counteracted by artificial respiration and mechanical respirators. Hypertension and circulatory collapse may be counteracted by use of intravenous fluids, plasma, or concentrated albumin and vasopressor agent such as norepinephrine. Epinephrine should not be used. In case of severe extrapyramidal reactions, antiparkinson medication should be administered.

Storage Condition

Store in a dry place (below 30°C). Keep container tightly closed. Protect from light.

Pack Size

Plastic container containing 30 tablets.

Product Registration Number

MAL19861364AZ

Further information can be obtained from your doctor or pharmacist.

ROYCE®

Product Holder / Manufactured by:
ROYCE PHARMA MFG SDN BHD 659435-X
PT 1663, Nilai Industrial Estate,
71800 Nilai, Negeri Sembilan, Malaysia.

Revision date: 050517

ROYCE®

Haloperidol Tablet 1.5mg

Presentation

Round, yellow coloured, flat face, bevel edge tablet with 7.9 mm diameter and breaking line on one side and plain on the other side.

Content:

Each tablet contains: Haloperidol 1.5mg.

Indication

Haloperidol is indicated for use in the management for manifestation of psychotic disorders. Haloperidol is indicated for the control of tics and vocal utterances of Tourette's disorder in children and adults. Haloperidol is effective for the treatment of severe behavior problems in children of combative, explosive hyperexcitability (WHICH CANNOT BE ACCOUNTED FOR BY IMMEDIATE PROVOCATION).
Haloperidol is effective in the short-term treatment of hyperactive children who show excessive motor activity with accompanying conduct disorders consisting of some or all of the following symptoms: impulsivity, difficulty sustaining attention, aggresivity, mood liability and poor frustration tolerance.

Dosage and administration

There is considerable variation from patient to patient in the amount of medication required for treatment. As with all neuroleptic drugs dosage should be individualized according to the needs response of each patient. Dosage adjustments, either upward or downward, should be carried out as rapidly as practicable to achieve optimum therapeutic control.

To determine the initial dosage, consideration should be given to the patient's age, severity of illness, previous response to other neuroleptic drugs and any concomitant medication of disease state. Children, debilitated or geriatric patients, as well as those with a history of adverse reactions to neuroleptic drugs may require less Haloperidol. The optimal response in such patients is usually obtained with more gradual dosage adjustments and at lower dosage levels, as recommended below. Clinical experience suggests following recommendations:

Oral administration
Initial Dosage Range
Adults: Moderate Symptomatology – 0.5 mg to 2.0 mg b.i.d or t.i.d
Severe Symptomatology – 3.0 mg to 5.0 mg b.i.d or t.i.d

To achieve prompt control, higher doses may be required in some cases.

Geriatric or Debilitated Patients – 0.5 mg to 2.0 mg b.i.d or t.i.d
Chronic or Resistant Patients – 3.0 mg to 5.0 mg b.i.d or t.i.d

Patients who remains severely distributed or inadequately controlled may require dosage adjustment. Daily dosages up to 100 mg may be necessary in some cases to achieve an optimal response. Infrequently, Haloperidol has been use in doses above 100 mg for severely resistant patients; however, the limited clinical usage has not demonstrated the safety of prolonged administration of such doses.

Children: The following recommendations apply to children between the ages 3 and 12 years (weight range 15 to 40 kg). Haloperidol is not intended for children under 3 years old. Therapy should begin at the lowest dose possible (0.5 mg per day). If required, the dose should be increased by an increment of 0.5 mg at 5 to 7 day intervals until the desired therapeutic effect is obtained (as listed below). The total dose may be divided, to be given b.i.d or t.i.d.

Psychotic Disorders - 0.05 mg/kg/day to 0.15 mg/kg/day.
Non-Psychotic Behavior – 0.05 mg/kg/day to 0.15 mg/kg/day.

Disorders and Tourette's Disorders
Severely distributed psychotics children may require higher doses. In severely distributed, non-psychotic children or in hyperactive children with accompanying conduct disorders, it should be noted that since these behaviors may be short-lived, short-term administrations of Haloperidol may suffice. There is no evidence establishing a maximum effective dosage. There is little evidence that behavior improvement is further enhanced in dosage beyond 6 mg per day.

Maintenance Dosage; Upon achieving a satisfactory therapeutic response, dosage should then be gradually reduced to the lowest effective maintenance level.

Pharmacological Information
The precise mechanism of action has not been clearly established.

Haloperidol is readily absorbed from the gastro-intestinal tract. It is metabolised in the liver and is excreted in the urine and faeces; there is evidence of enterohepatic recycling. Owing to the firs -pass effect of metabolism in the liver plasma concentrations following oral administration are lower than those following intramascular administration. Moreover, there is wide inter subject variation in plasma concentrations of haloperidol, but in practice, no simple correlation has been found between plasma concentrations of haloperidol and its therapeutic effect. Paths of metabolism and its therapeutic include oxidative N-dealkylayion. Haloperidol has been reported to have a plasma half-life ranging from about 13 to nearly 40 hours; its plasma half-life is prolonged during the night. Haloperidol is very extensively bound to plasma proteins; it is widely in the body and crosses the blood-brain barrier.

Contarindication
Haloperidol is contraindicated is severe toxic central nervous system depression or comatose states from any cause and individuals who are hypersensitive to this drug or have Parkinson's disease.

Warning and Precautions
This drug should be used during pregnancy or in women likely to become pregnant only if the benefit clearly justifies a potential risk to the fetus. Infants should not be nursed during treatment.

Combined use of Haloperidol and Lithium: An encephalopathic syndrome (characterized by weakness, lethargy, fever, tremulousness and confusion, extrapyramidal symptoms, leukocytosis, elevated serum enzymes, BUN, and FBS) followed by irreversible brain damage has occurred in a few patients treated with lithium plus Haloperidol. A causal relationship between these events and the concomitant administration of lithium and Haloperidol has not been established; however, patients receiving such combined therapy should be monitored closely for early evidence of neurological toxicity and treatment discontinued promptly if such signs appear.

General: A number of cases of bronchopneumonia, some fatal, have followed the use of major tranquilizers, including Haloperidol. It has been postulated that lethargy and decreased sensation of thirst due to central inhibition may lead to dehydration, hemoconcentration and reduced pulmonary ventilation.

Therefore, if the above signs and symptoms appear, especially in the elderly, the physician should institute remedial therapy promptly. Although not reported with Haloperidol, decreased serum cholesterol and/or cutaneous and ocular changes have been reported in patients receiving chemically related drugs.

Haloperidol may impair the mental and/or physical abilities required for the performance of hazardous tasks such as operating machinery or driving a motor vehicle. The ambulatory patient should be warned accordingly. The use of alcohol with this drug should be avoided due to possible additive effects and hypotension.

Haloperidol should be administered cautiously to patients:
- with severe cardiovascular disorders, because of the possibility of transient hypotension and/or precipitation of anginal pain. Should hypotension occur and a vasopressor be required epinephrine should not be use since Haloperidol may block its vasopressor activity and paradoxical further lowering of the blood pressure may occur.
- receiving anticonvulsant medication, because Haloperidol may lower the convulsive threshold. Adequate anticonvulsant therapy should be maintained concomitantly.
- with known allergies, or with a history of allergic reactions to drugs. Occurred with the effects of one anticoagulant (phenindione).

If concomitant antiparkinson medication is required, it may have to be continued after Haloperidol is discontinued because of the difference in excretion rates. If both are discontinued simultaneously, extrapyramidal symptoms may occur. The physician should keep in mind the possible increase in intraocular pressure when anticholinergic drugs, including antiparkinson agent, are administered concomitantly with Haloperidol.

When Haloperidol is used to control mania in cyclic disorders, there may be a rapid mood swing to depression.

Severe neurotoxicity (rigidity, inability to walk or talk) may occur in patients with thyrotoxicosis who are also receiving antipsychotic medication, including Haloperidol.

Neuroleptic drugs elevate prolactin levels: the elevation persists during chronic administrations.

Side Effects
CNS Effects Estrapyramidal Reactions-Neuromuscular (extrapyramidal) reactions during the administration of Haloperidol have been reported frequently, often during the first few days of treatment. In most patients, these reactions involved Parkinson-like symptoms which, when first observed, were usually mild to moderately severe and usually reversible. Other types of neuromuscular reactions (motor restlessness dystonia, akathisia, hyperreflexia opisthotonos, oculogyric crises) have been reported far less frequently, but were often more severe. Severe extrapyramidal reactions have been reported to occur at relatively low doses. Generally the occurrence and severity of most extrapyrimidal symptoms are dose related since they occur at relatively high doses and have been shown to disappear or become less severe when the dose is reduced. Administration of antiparkinson drugs such as benztropine mesylate U.S.P. or trihexyphenidyl hydrochloride
U.S.P may be required for control of such reactions. It should be noted that persistent extrapyramidal reactions have been reported and that the drug may have to be discontinued in such cases.

Withdrawal Emergent Neurological Signs-Generally, patients receiving short term therapy experience no problems with abrupt discontinuation of antipsychotic drugs. However some patients on maintenance treatment experience transient dyskinetic signs after abrupt withdrawal. In certain of these cases the dyskinetic movements are indistinguishable from the syndrome described below under "persistent Tardive Dyskinesia" except for duration. It is not known whether gradual withdrawal of antipsychotic drugs will reduce the rate or occurrence of withdrawal emergent neurological signs but until further evidence becomes available, it seems reasonable to gradually withdraw use of Haloperidol.

Persistent Tardive Dyskinesia-As with all antipsychotic agents Haloperidol has been associated with persistent dyskinesia. Tardive dyskinesia may appear in some patients on long-term therapy or may occur after drug therapy has been discontinued.

The risk appears to be greater in elderly patients on high dose therapy, especially females. The symptoms are persistent and some patients appear irreversible. The syndrome is characterized by rhythmical involuntary movements of tongue, face, mouth or jaw (e.g., protrusion of tongue, puffing of cheeks, puckering of mouth, chewing movements). Sometimes these may be accompanied by involuntary movements of extremities.